

cs224n a2 mzx

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(a)

$$- \sum_{w \in \text{Vocab}} y_w \log(\hat{y}_w) = -(y_o \log(\hat{y}_o) + \sum_{w \in \text{Vocab}, w \neq o} y_w \log(\hat{y}_w)) = -\log(\hat{y}_o),$$

where $y_o = 1$, $y_w = 0$ for $w \neq o$.

(b)

$$\begin{aligned}\frac{\partial J_{\text{naive-softmax}}(v_c, o, U)}{\partial v_c} &= -\frac{\partial}{\partial v_c} \log \mathbb{P}(O = o \mid C = c) \\&= -\frac{\partial}{\partial v_c} \log \frac{\exp(u_o^\top v_c)}{\sum_{w \in \text{Vocab}} \exp(u_w^\top v_c)} \\&= -\frac{\partial}{\partial v_c} \left[\log \exp(u_o^\top v_c) - \log \sum_{w \in \text{Vocab}} \exp(u_w^\top v_c) \right] \\&= -\frac{\partial}{\partial v_c} \left[u_o^\top v_c - \log \sum_{w \in \text{Vocab}} \exp(u_w^\top v_c) \right] \\&= -u_o + \frac{\partial}{\partial v_c} \left[\log \sum_{w \in \text{Vocab}} \exp(u_w^\top v_c) \right] \\&= -u_o + \frac{1}{\sum_{w \in \text{Vocab}} \exp(u_w^\top v_c)} \frac{\partial}{\partial v_c} \left[\sum_{w \in \text{Vocab}} \exp(u_w^\top v_c) \right] \\&= -u_o + \frac{1}{\sum_{w \in \text{Vocab}} \exp(u_w^\top v_c)} \left[\sum_{w \in \text{Vocab}} \exp(u_w^\top v_c) \frac{\partial}{\partial v_c} (u_w^\top v_c) \right] \\&= -u_o + \frac{1}{\sum_{w \in \text{Vocab}} \exp(u_w^\top v_c)} \left[\sum_{w \in \text{Vocab}} \exp(u_w^\top v_c) u_w \right] \\&= -u_o + \sum_{w \in \text{Vocab}} \frac{\exp(u_w^\top v_c) u_w}{\sum_{x \in \text{Vocab}} \exp(u_x^\top v_c)} \\&= -u_o + \sum_{w \in \text{Vocab}} \mathbb{P}(O = w \mid C = c) u_w \\&= -u_o + \sum_{w \in \text{Vocab}} \hat{y}_w u_w \\&= -y_o u_o + \sum_{w \in \text{Vocab}} \hat{y}_w u_w \\&= -\sum_{x \in \text{Vocab}} y_x u_x + \sum_{w \in \text{Vocab}} \hat{y}_w u_w \\&= \sum_{x \in \text{Vocab}} (\hat{y}_x - y_x) u_x \\&= U(\hat{y} - y)\end{aligned}$$

(c)

Case 1: $w \neq o$

$$\begin{aligned}
\frac{\partial J_{\text{naive-softmax}}(v_c, o, U)}{\partial u_w} &= -\frac{\partial}{\partial u_w} \left[u_o^\top v_c - \log \sum_{x \in \text{Vocab}} \exp(u_x^\top v_c) \right] \\
&= \frac{\partial}{\partial u_w} \left[\log \sum_{x \in \text{Vocab}} \exp(u_x^\top v_c) \right] \\
&= \frac{1}{\sum_{x \in \text{Vocab}} \exp(u_x^\top v_c)} \frac{\partial}{\partial u_w} \left[\sum_{x \in \text{Vocab}} \exp(u_x^\top v_c) \right] \\
&= \frac{1}{\sum_{x \in \text{Vocab}} \exp(u_x^\top v_c)} \left[\sum_{x \in \text{Vocab}} \frac{\partial}{\partial u_w} \exp(u_x^\top v_c) \right] \\
&= \frac{1}{\sum_{x \in \text{Vocab}} \exp(u_x^\top v_c)} \left[\sum_{x \in \text{Vocab}} \exp(u_x^\top v_c) \frac{\partial}{\partial u_w} (u_x^\top v_c) \right] \\
&= \frac{1}{\sum_{x \in \text{Vocab}} \exp(u_x^\top v_c)} \exp(u_w^\top v_c) \frac{\partial}{\partial u_w} (u_w^\top v_c) \\
&= \frac{\exp(u_w^\top v_c)}{\sum_{x \in \text{Vocab}} \exp(u_x^\top v_c)} v_c \\
&= \hat{y}_w v_c
\end{aligned}$$

Case 2: $w = o$

$$\begin{aligned}
\frac{\partial J_{\text{naive-softmax}}(v_c, o, U)}{\partial u_w} &= -\frac{\partial}{\partial u_w} \left[u_o^\top v_c - \log \sum_{x \in \text{Vocab}} \exp(u_x^\top v_c) \right] \\
&= -\frac{\partial}{\partial u_w} (u_o^\top v_c) + \hat{y}_w v_c \\
&= -v_c + \hat{y}_w v_c \\
&= (\hat{y}_w - 1) v_c
\end{aligned}$$

Thus,

$$\frac{\partial J_{\text{naive-softmax}}}{\partial u_w} = \begin{cases} (\hat{y}_o - 1) v_c, & w = o \\ \hat{y}_w v_c, & w \neq o \end{cases}$$

Equivalently, since $y_o = 1$ and $y_w = 0$ ($w \neq o$), this can be written as:

$$\frac{\partial J_{\text{naive-softmax}}}{\partial u_w} = (\hat{y}_w - y_w) v_c.$$

(d)

$$\begin{aligned}\frac{d}{dx}\sigma(x) &= \frac{d}{dx} \frac{1}{1 + \exp(-x)} \\ &= -(1 + \exp(-x))^{-2} \frac{d}{dx}(1 + \exp(-x)) \\ &= -(1 + \exp(-x))^{-2} \frac{d}{dx}(\exp(-x)) \\ &= -(1 + \exp(-x))^{-2} \exp(-x) \frac{d}{dx}(-x) \\ &= \frac{\exp(-x)}{(1 + \exp(-x))^2} \\ &= \frac{1}{1 + \exp(-x)} \cdot \frac{\exp(-x)}{1 + \exp(-x)} \\ &= \sigma(x) \cdot (1 - \sigma(x))\end{aligned}$$

(e)

$$\begin{aligned}\frac{\partial}{\partial z} \log \sigma(z) &= \frac{1}{\sigma(z)} \cdot \sigma(z)(1 - \sigma(z)) = 1 - \sigma(z) \\ \frac{\partial}{\partial z} \log \sigma(-z) &= \frac{1}{\sigma(-z)} \cdot \sigma(-z)(1 - \sigma(-z)) \cdot (-1) = \sigma(-z) - 1 = -\sigma(z)\end{aligned}$$

(1) $\partial J / \partial v_c$:

$$\begin{aligned}\frac{\partial}{\partial v_c} J_{\text{neg-sample}}(v_c, o, U) &= -\frac{\partial}{\partial v_c} \left[\log(\sigma(u_o^\top v_c)) + \sum_{k=1}^K \log(\sigma(-u_k^\top v_c)) \right] \\ &= -\left[[1 - \sigma(u_o^\top v_c)] \frac{\partial}{\partial v_c} u_o^\top v_c + \frac{\partial}{\partial v_c} \sum_{k=1}^K \log(\sigma(-u_k^\top v_c)) \right] \\ &= [\sigma(u_o^\top v_c) - 1] u_o - \sum_{k=1}^K \frac{\partial}{\partial v_c} \log(\sigma(-u_k^\top v_c)) \\ &= [\sigma(u_o^\top v_c) - 1] u_o - \sum_{k=1}^K -\sigma(u_k^\top v_c) \frac{\partial}{\partial v_c} (u_k^\top v_c) \\ &= [\sigma(u_o^\top v_c) - 1] u_o + \sum_{k=1}^K \sigma(u_k^\top v_c) u_k\end{aligned}$$

(2) $\partial J / \partial u_o$:

$$\begin{aligned}\frac{\partial}{\partial u_o} J_{\text{neg-sample}}(v_c, o, U) &= -\frac{\partial}{\partial u_o} \left[\log(\sigma(u_o^\top v_c)) + \sum_{k=1}^K \log(\sigma(-u_k^\top v_c)) \right] \\ &= -\frac{\partial}{\partial u_o} \log(\sigma(u_o^\top v_c)) \\ &= [\sigma(u_o^\top v_c) - 1] v_c\end{aligned}$$

(3) $\partial J / \partial u_k$ (for some $k \in [1, K]$):

$$\begin{aligned}\frac{\partial}{\partial u_k} J_{\text{neg-sample}}(v_c, o, U) &= -\frac{\partial}{\partial u_k} \left[\log(\sigma(u_o^\top v_c)) + \sum_{k'=1}^K \log(\sigma(-u_{k'}^\top v_c)) \right] \\ &= -\frac{\partial}{\partial u_k} \sum_{k'=1}^K \log(\sigma(-u_{k'}^\top v_c))\end{aligned}$$

Notice that $\frac{\partial(u_{k'}^\top v_c)}{\partial u_k} = v_c$ when $k' = k$, otherwise it is 0.

$$\begin{aligned}\frac{\partial J}{\partial u_k} &= -[-\sigma(u_k^\top v_c) \cdot v_c] \\ &= \sigma(u_k^\top v_c) v_c\end{aligned}$$

negative sampling loss only involves $K + 1$ dot products. naive softmax requires summing over the entire vocabulary

(f)

(1)

$$\frac{\partial J_{\text{skip-gram}}(v_c, w_{t-m}, \dots, w_{t+m}, U)}{\partial U} = \sum_{-m \leq j \leq m, j \neq 0} \frac{\partial J(v_c, w_{t+j}, U)}{\partial U}$$

(2)

$$\frac{\partial J_{\text{skip-gram}}(v_c, w_{t-m}, \dots, w_{t+m}, U)}{\partial v_c} = \sum_{-m \leq j \leq m, j \neq 0} \frac{\partial J(v_c, w_{t+j}, U)}{\partial v_c}$$

(3)

$$\frac{\partial J_{\text{skip-gram}}(v_c, w_{t-m}, \dots, w_{t+m}, U)}{\partial v_w} = \sum_{-m \leq j \leq m, j \neq 0} \frac{\partial J(v_c, w_{t+j}, U)}{\partial v_w} = \sum_{-m \leq j \leq m, j \neq 0} 0 = 0$$